Chapter 6, Problem 27P

Problem

Given that four 10-μF capacitors can be connected in series and in parallel, find the minimum and maximum values that can be obtained by such series/parallel combinations.

Step-by-step solution

Step 1 of 3

Refer to section 6.3 in the textbook.

Consider the four $10\ \mu\text{F}$ capacitors.

Step 2 of 3

Minimum capacitance can be obtained with the series combination of four capacitors.

Consider the series combination.

Determine the equivalent capacitance of the four $4\ \mu\text{F}$ capacitors connected in series.

$$\frac{1}{C_{eq}}=\frac{1}{C_1}+\frac{1}{C_2}+\frac{1}{C_3}+\frac{1}{C_4}$$

Substitute $C_1=C_2=C_3=C_4=4\ \mu\text{F}$.

$$\begin{aligned}\frac{1}{C_{eq}}&=\frac{1}{10\ \mu\text{F}}+\frac{1}{10\ \mu\text{F}}+\frac{1}{10\ \mu\text{F}}+\frac{1}{10\ \mu\text{F}}\\&=\frac{(1+1+1+1)}{10\ \mu\text{F}}\\&=\frac{4}{10\ \mu\text{F}}\\&=\frac{2}{5\ \mu\text{F}}\end{aligned}$$

$$\begin{aligned}C_{eq}&=\frac{5\ \mu\text{F}}{2}\\&=2.5\ \mu\text{F}\end{aligned}$$

Thus, the minimum capacitance of the combination is $\boxed{2.5\ \mu\text{F}}$.

Step 3 of 3

Maximum capacitance can be obtained with the parallel combination of four capacitors.

Consider the parallel combination.

Determine the equivalent capacitance of the four $4\ \mu\text{F}$ capacitors connected in parallel.

$$C_{eq}=C_1+C_2+C_3+C_4$$

Substitute $C_1=C_2=C_3=C_4=4\ \mu\text{F}$.

$$\begin{aligned}C_{eq}&=10\ \mu\text{F}+10\ \mu\text{F}+10\ \mu\text{F}+10\ \mu\text{F}\\&=40\ \mu\text{F}\end{aligned}$$

Thus, the maximum capacitance of the combination is $\boxed{40\ \mu\text{F}}$.